

# Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LAILA | Context: Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

LAILA's task taxonomy is explicitly limited to two genres (explanatory and persuasive) [\[Q89\]](#) and operationalizes essay scoring through the QUTC CAST seven-trait rubric [\[Q33, Q34\]](#), a Qatari-curriculum artifact. For a pan-Arab teacher-facing deployment, this presents two HIGH-priority IO problems flagged in the elicitation: (1) Egypt's secondary Arabic assessment is holistic and rubric-free with no documented multi-trait analogue [\[WEB-5\]](#); the seven-trait structure does not map onto Egyptian classroom practice. (2) Narrative and descriptive genres, which are present in many Arab national curricula, are absent from the benchmark and are explicitly noted as a limitation by the authors [\[Q89\]](#). The benchmark also frames AES purely as score prediction (regression / sequential generation of trait labels) [\[Q147, Q181\]](#) and does not include any task category for confidence estimation or rationale generation, both of which the deployment requires. The two evaluation setups (prompt-specific and cross-prompt) [\[Q14\]](#) are methodologically appropriate but do not address cross-national generalization. No public rubric documentation was found for Saudi Arabia, Jordan, Lebanon, or Morocco that would allow the seven-trait mapping to be confirmed [\[WEB-7, WEB-10\]](#). The strengths are real (multi-trait coverage, two evaluation setups), but the omissions are central to deployment validity.

| ID                     | Evidence                                                                                                                                                                                                                                                       |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q89</a>    | "Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing." (p.10)                                            |
| <a href="#">Q33</a>    | "All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)                                                                                               |
| <a href="#">Q34</a>    | "The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)                                                                                    |
| <a href="#">Q147</a>   | "The AES task is formulated as a text generation problem, where the model predicts the trait scores sequentially." (p.16)                                                                                                                                      |
| <a href="#">Q181</a>   | "In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)                                                                                                             |
| <a href="#">Q14</a>    | "We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt." (p.2)                                                                                                                                            |
| <a href="#">WEB-5</a>  | Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators ( <a href="https://grokipedia.com/page/academic_grading_in_egypt">https://grokipedia.com/page/academic_grading_in_egypt</a> ) |
| <a href="#">WEB-7</a>  | Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics ( <a href="https://etec.gov.sa/en/centers/qiyas">https://etec.gov.sa/en/centers/qiyas</a> )                                                          |
| <a href="#">WEB-10</a> | Jordan's Tawjihi Arabic exam emphasizes literary comprehension and grammar over trait-scored free composition ( <a href="https://en.wikipedia.org/wiki/Tawjihi">https://en.wikipedia.org/wiki/Tawjihi</a> )                                                    |

## Strengths

- Provides fine-grained seven-trait coverage (REL, ORG, VOC, STY, DEV, MEC, GRA) supporting analytic rather than only holistic scoring [\[Q34, Q115\]](#)
- Includes both prompt-specific and cross-prompt evaluation setups, with cross-prompt being the more deployment-realistic case [\[Q14, Q72\]](#)
- Evaluates a diverse model taxonomy (feature-based, encoder-based, Arabic-centric LLMs in zero/few-shot) [\[Q48, Q51\]](#), giving teachers a meaningful baseline landscape

| ID                   | Evidence                                                                                                                                                                                                                                                            |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q34</a>  | "The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)                                                                                         |
| <a href="#">Q115</a> | "This rubric guided scoring across 7 writing traits: relevance (REL), organization (ORG), vocabulary (VOC), style (STY), development (DEV), mechanics (MEC), and grammar (GRA)." (p.12)                                                                             |
| <a href="#">Q14</a>  | "We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt." (p.2)                                                                                                                                                 |
| <a href="#">Q72</a>  | "We note that the applicability of the prompt-specific setup is relatively limited due to reliance on target prompt essays that are labeled, which is often unavailable in practice." (p.9)                                                                         |
| <a href="#">Q48</a>  | "The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly." (p.7) |
| <a href="#">Q51</a>  | "For LLMs, we evaluated three Arabic-centric models: ALLaM (Bari et al., 2025), Command-R7B-Arabic (R7B) (Alnumay et al., 2025), and Fanar (Team et al., 2025), under zero-shot and few-shot (5-shot) settings." (p.7)                                              |

## Checklist

### Checklist item definitions:

| ID                   | Assessment Question                                            |
|----------------------|----------------------------------------------------------------|
| <a href="#">IO-1</a> | Identify test case categories required for regional deployment |
| <a href="#">IO-2</a> | Check if taxonomy omits regionally relevant categories         |
| <a href="#">IO-3</a> | Check if taxonomy includes irrelevant categories               |
| <a href="#">IO-4</a> | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** Required deployment categories include: (a) trait-based analytic scoring for Gulf-style rubric users, (b) holistic-only scoring suited to Egyptian practice, (c) narrative and descriptive genre scoring, and (d) confidence/rationale outputs. LAILA covers (a) explicitly [\[Q34\]](#) but does not cover (b), (c), or (d).

**IO-2:** Yes. The taxonomy omits narrative and descriptive genres [\[Q89\]](#), omits a holistic-only scoring mode aligned with Egyptian practice [\[WEB-5\]](#), and omits confidence/rationale categories that the deployment requires [\[elicitation Q3\]](#). No mapping to Tawjihi (Jordan) or Qiyas (Saudi Arabia) frameworks is documented [\[WEB-7, WEB-10\]](#).

**IO-3:** No clearly irrelevant categories were identified — all seven CAST traits are plausibly meaningful for MSA writing, even if not all are weighted equivalently across national curricula. The pan-Arab MSA standards assumption from elicitation supports surface relevance of the trait set.

**IO-4:** Documented gaps that harm content validity: (1) genre ontology restricted to explanatory/persuasive [\[Q89\]](#); (2) no holistic-only or rubric-free evaluation mode for Egyptian-style use [\[WEB-5\]](#); (3) no task category for uncertainty or rationale generation [\[Q181, Q182\]](#); (4) no cross-national rubric mapping between CAST traits and Saudi/Jordanian/Lebanese/Moroccan frameworks [\[WEB-7, WEB-10\]](#).

| ID                     | Evidence                                                                                                                                                                                                                                                       |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q34</a>    | "The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)                                                                                    |
| <a href="#">WEB-5</a>  | Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators ( <a href="https://grokipedia.com/page/academic_grading_in_egypt">https://grokipedia.com/page/academic_grading_in_egypt</a> ) |
| <a href="#">Q89</a>    | "Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing." (p.10)                                            |
| <a href="#">WEB-7</a>  | Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics ( <a href="https://etec.gov.sa/en/centers/qiyas">https://etec.gov.sa/en/centers/qiyas</a> )                                                          |
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|      |                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q3   | "Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1) |
| Q181 | "In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)                              |
| Q182 | "The holistic score was computed as the sum of the individual trait scores." (p.18)                                                                                             |

### Information Gaps

- Genre distribution (narrative/descriptive/argumentative/explanatory) in G10–12 Arabic curricula across Egypt, Saudi Arabia, Jordan, Lebanon, Morocco
- Whether national curricula in target countries use any analytic trait dimensions that would map onto CAST

### Requires Expert Verification

- Mapping of CAST traits to Lebanese CRDP and Moroccan national Arabic writing criteria
- Whether Saudi MOE classroom Arabic assessment uses any internal trait-based rubric beyond Qiyas

### Remediation

**Gap 1: Seven-trait CAST taxonomy does not map onto Egyptian holistic grading practice and lacks narrative/descriptive genre coverage**

**Recommendation:** Add a holistic-only evaluation track suitable for Egyptian-style assessment, and either commission a narrative/descriptive prompt extension or document explicitly that the system is restricted to explanatory/persuasive genres in deployment guidelines.

**Gap 2: Cross-prompt generalization performance is substantially weaker than prompt-specific performance [Q71], yet cross-prompt is the realistic deployment setting [Q72]**

**Recommendation:** Anchor deployment evaluation primarily on cross-prompt results rather than prompt-specific results, and communicate this performance ceiling clearly to teacher users so they understand the system's reliability limits on novel prompts.

| ID  | Evidence                                                                                                                                                                                    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q71 | "However, its cross-prompt performance drops significantly (average QWK: 0.549), underscoring its ability to capture prompt-dependent patterns rather than generalizable features." (p.9)   |
| Q72 | "We note that the applicability of the prompt-specific setup is relatively limited due to reliance on target prompt essays that are labeled, which is often unavailable in practice." (p.9) |